Balanced Stieltjes Moments

Bernoulli Closure, Harmonic Filters,
and Polylogarithmic Resonances

A research continuation prepared for Vladimir Reshetnikov

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Abstract

We construct a binomially balanced family of generalized Stieltjes functions whose first several inverse-power tails cancel identically. Its Mellin transform is a product of a Hurwitz zeta function, the reflection factor $\pi \csc(\pi u)$, and one Gamma-quotient coefficient. Every convergent positive-integer moment consequently reduces to a Bernoulli polynomial times a finite rational combination of ordinary zeta values. At a zero of the relevant Bernoulli polynomial, one additional moment converges; its value is instead a Hurwitz-zeta order derivative, or equivalently an explicit combination of polylogarithm order derivatives.

We prove uniqueness of the normalized homogeneous balancing, a necessary and sufficient finite-shift cancellation criterion, and a two-variable formula for all logarithmic moments of filtered Stieltjes constants. The last formula has coefficients generated by finite harmonic sums and ordinary zeta values. It supplies an exact distinction between elementary and complete homogeneous harmonic coordinates. Shift derivatives and anchored primitives give further convergent identities, including an all-order centered vanishing family and an odd-zeta evaluation at the first exceptional centered moment.

The classical beta integral, Euler–Maclaurin formula, and Hurwitz functional equation are credited inputs. The specific all-order organization is offered as a proved research contribution for independent review, not as a certified literature-priority claim. Existing Gaussian S_6 and revised S_8 conjectures are not resolved here.

Keywords. Generalized Stieltjes constants; Mellin transforms; Bernoulli polynomials; Gamma quotients; harmonic sums; polylogarithm order derivatives; ordinary convergent integrals.

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1 Scope, motivation, and principal results

The ProveIt manuscript develops bridges among polylogarithms, Hurwitz jets, Stieltjes calculus, and Gamma correlations. The inspected harmonic-generating notes explicitly distinguish their known constant terms from genuinely mixed harmonic–Stieltjes coefficients [8, 9]. The inspected Mellin continuation integrates polylogarithms in their argument and obtains Hurwitz and Gamma expressions [10]. We study a different integral: a Mellin transform in the *positive translation variable* x of Stieltjes functions at $x + a$.

A term such as $\gamma_q(x + a)$ by itself has a growing logarithmic asymptotic expansion, so it has no positive-integer moment on $(0, \infty)$. The combination

$$\mathcal{C}_q(x, a) = \frac{\log^{q+1} x}{q+1} + \sum_{j=0}^{q} (-1)^j \binom{q}{j} \gamma_j(x + a) \log^{q-j} x \tag{1.1}$$

removes not merely that growth but all inverse powers $x^{-1}, \dots, x^{-q}$. The resulting ordinary integral satisfies

$$\int_0^\infty x^{u-1} \mathcal{C}_q(x, a) \, \mathrm{d}x = \frac{\pi}{\sin \pi u} \zeta(1 - u, a) Q_q(u), \qquad 0 < \operatorname{Re} u < q + 1, \tag{1.2}$$

where

$$\sum_{q \geq 0} \frac{Q_q(u)}{q!} z^q = \frac{\Gamma(u + z)}{\Gamma(u)\Gamma(1 + z)}. \tag{1.3}$$

Interior integer singularities on the right are removable. Their removal is an essential part of the theorem, not an informal rule for multiplying zero by infinity.

For $q \geq m \geq 1$, let

$$\kappa_{m,q} = \frac{(-1)^{m+q} q!}{m} \sum_{j=0}^{m-1} (-1)^j e_j(1, 1/2, \dots, 1/(m-1)) \zeta(q + 1 - j). \tag{1.4}$$

Here $e_0 = 1$ and an empty positive-degree elementary symmetric sum is zero. The complete ordinary moment is

$$\boxed{\int_0^\infty x^{m-1} \mathcal{C}_q(x, a) \, \mathrm{d}x = \kappa_{m,q} B_m(a).} \tag{1.5}$$

For example,

$$\int_0^\infty \mathcal{C}_1(x, a) \, \mathrm{d}x = (a - \tfrac{1}{2}) \zeta(2), \tag{1.6}$$

$$\int_0^\infty x \mathcal{C}_2(x, a) \, \mathrm{d}x = \big(\zeta(3) - \zeta(2)\big)(a^2 - a + \tfrac{1}{6}). \tag{1.7}$$

A distinct cancellation occurs at a Bernoulli root:

$$B_m(a) = 0 \quad \Longrightarrow \quad \int_0^\infty x^{m-1} \mathcal{C}_{m-1}(x, a) \, \mathrm{d}x = (-1)^{m-1} \zeta^{[1]}(1 - m, a). \tag{1.8}$$

In particular,

$$\boxed{\int_0^\infty x^2 \mathcal{C}_2(x, \tfrac{1}{2}) \, \mathrm{d}x = \frac{3\zeta(3)}{16\pi^2}.} \tag{1.9}$$

Superscript $[r]$ always denotes differentiation in the *order* of zeta or polylogarithm, whereas parentheses on $\psi^{(r)}$ and $\gamma_n^{(r)}$ denote differentiation in their argument.

The article then classifies which finite combinations of shifts admit additional moments and gives a full mixed logarithmic formula. The classification concerns exact analytic convergence and reduction in specified coordinates. It is not an arithmetic-independence theorem.

1.1 What is and is not claimed

The beta transform underlying (1.2), the Gamma expansions in (1.3), Bernoulli special values, and the Hurwitz Fourier formula are classical [1, 2, 6]. Our contribution is the combined cancellation theorem, its uniqueness within a precisely stated ansatz, the two different resonance mechanisms, and the finite-shift and mixed-moment consequences proved below. A focused source comparison did not establish an earlier identical all-order formulation. That observation is not an exhaustive priority search, particularly for the unexpanded incoming archives.

No theorem below evaluates the unresolved coefficients $R_{r,k}(a)$ of the repository's harmonic Barnes–Hurwitz family. Our $\gamma_n(x+a)$ are ordinary generalized Stieltjes constants, not those higher harmonic constants. No result is asserted for the Gaussian S_6 or revised S_8 period residuals. These distinctions avoid silently changing the problem whose solution is claimed.

2 Balanced Stieltjes functions and cancellation

2.1 Conventions and the holomorphic generator

Throughout the main theorems, $a > 0$, $x > 0$, and $\log x$ is real. The normalization is

$$\zeta(1+z,a) = \frac{1}{z} + \sum_{n=0}^{\infty} \frac{(-1)^n \gamma_n(a)}{n!} z^n, \qquad \gamma_0(a) = -\psi(a). \tag{2.1}$$

Bernoulli polynomials have the standard normalization [5] and generating function

$$\frac{te^{at}}{e^t - 1} = \sum_{k=0}^{\infty} B_k(a) \frac{t^k}{k!}, \qquad B_1(a) = a - \frac{1}{2}. \tag{2.2}$$

Put $P_n(z) = (1+z)_n$, with $P_0 = 1$. Define

$$F(z;x,a) = x^z \zeta(1+z, x+a) - \frac{1}{z}. \tag{2.3}$$

The singularity at $z = 0$ is removable. In fact F is entire in z for each $x, a > 0$, since the only order pole of Hurwitz zeta has been removed. Direct multiplication of (2.1) gives

$$\mathcal{C}_q(x,a) = \partial_z^q F(z;x,a)\big|_{z=0}, \qquad F(z;x,a) = \sum_{q \geq 0} \mathcal{C}_q(x,a) \frac{z^q}{q!}. \tag{2.4}$$

The first functions are

$$\mathcal{C}_0(x,a) = \log x - \psi(x+a), \tag{2.5}$$

$$\mathcal{C}_1(x,a) = -\psi(x+a)\log x - \gamma_1(x+a) + \tfrac{1}{2}\log^2 x, \tag{2.6}$$

$$\mathcal{C}_2(x,a) = -\psi(x+a)\log^2 x - 2\gamma_1(x+a)\log x + \gamma_2(x+a) + \tfrac{1}{3}\log^3 x. \tag{2.7}$$

Lemma 2.1 (Normally uniform inverse-power expansion)**.** *For every integer $K \geq 1$, uniformly for z in a sufficiently small closed disc about zero and for a in a compact subset of $(0, \infty)$,*

$$F(z; x, a) = \sum_{k=1}^{K} \frac{(-1)^k B_k(a)}{k!} P_{k-1}(z) x^{-k} + O(x^{-K-1}) \qquad (x \longrightarrow \infty). \tag{2.8}$$

Consequently, for fixed $q \geq 0$,

$$\mathcal{C}_q(x, a) = \sum_{k=q+1}^{K} \frac{(-1)^k B_k(a)}{k!} P_{k-1}^{(q)}(0) x^{-k} + O(x^{-K-1}), \tag{2.9}$$

where the sum is empty when $K \leq q$. In particular,

$$\mathcal{C}_q(x, a) = \frac{(-1)^{q+1} B_{q+1}(a)}{q+1} x^{-q-1} + O(x^{-q-2}). \tag{2.10}$$

At zero, $\mathcal{C}_q(x, a) = O(1 + |\log x|^{q+1})$, uniformly on the same compact sets of a.

Proof. Euler–Maclaurin applied to the defining Hurwitz series, followed by expansion in a/x, gives the standard shifted asymptotic formula

$$\zeta(s, x + a) = \frac{x^{1-s}}{s-1} + \sum_{k=1}^{K} \frac{(-1)^k B_k(a)}{k!} (s)_{k-1} x^{1-s-k} + O(|x^{1-s}| x^{-K-1}).$$

It is enough to use more Euler–Maclaurin terms than K; their bounded periodic-Bernoulli remainder yields this estimate uniformly for s in a compact set near 1 after the leading pole is removed. These are the usual local uniformity properties of the Hurwitz asymptotic expansion [1, §25.11(iii),(xii)].

For completeness, the Bernoulli coefficient can be checked without assuming the shifted form. The leading integral, half-endpoint term, and the rth even Euler–Maclaurin term contribute at x^{-k}, after multiplication by x^{s-1}, a common factor $(s)_{k-1}$ times

$$\frac{(-1)^k a^k}{k!} + \frac{(-1)^{k-1} a^{k-1}}{2(k-1)!} + \sum_{r=1}^{\lfloor k/2 \rfloor} \frac{B_{2r} (-1)^{k-2r} a^{k-2r}}{(2r)!(k-2r)!}.$$

By (2.2), this is $(-1)^k B_k(a)/k!$. Set $s = 1 + z$ to obtain (2.8). Uniformity on a slightly larger z-disc permits Cauchy differentiation of its remainder, with the same inverse-power bound and no additional logarithm of x.

Since P_{k-1} has degree $k-1$, its qth derivative vanishes for $k \leq q$. Also $P_q^{(q)}(0) = q!$, giving (2.10). Finally, for x near zero all $\gamma_j(x + a)$ are uniformly bounded on compact positive a-intervals. Formula (1.1) proves the last estimate. □

Corollary 2.2 (Ordinary convergence strip)**.** *For each $q \geq 0$ the integral*

$$\mathcal{J}_q(u, a) = \int_0^\infty x^{u-1} \mathcal{C}_q(x, a) \, \mathrm{d}x \tag{2.11}$$

converges absolutely and is holomorphic in u on $0 < \operatorname{Re} u < q + 1$. It is locally uniformly convergent in $a > 0$. If $B_{q+1}(a) = 0$, it converges on the larger strip $0 < \operatorname{Re} u < q + 2$ for that fixed a.

Proof. Apply Lemma 2.1 at the two endpoints. On a compact substrip, additional powers of $\log x$ remain integrable, proving holomorphy. □

2.2 Why the binomial coefficients are forced

Theorem 2.3 (Uniqueness in the homogeneous logarithmic ansatz)**.** *Fix $q \geq 0$. Consider constants $d_0, \ldots, d_q, b$ with $d_0 = 1$ and*

$$D_q(x,a) = b \log^{q+1} x + \sum_{j=0}^{q} d_j \gamma_j(x+a) \log^{q-j} x. \tag{2.12}$$

Suppose $D_q(x,a) = O(x^{-q-1})$ as $x \to \infty$ for every a in some nonempty open positive interval. Then

$$d_j = (-1)^j \binom{q}{j}, \qquad b = \frac{1}{q+1}, \qquad D_q = \mathcal{C}_q. \tag{2.13}$$

Proof. For $q = 0$, cancellation of the leading logarithm forces $b = 1$. Let $q \geq 1$. Write $L = \log x$ and $d(t) = \sum_{j=0}^{q} d_j t^j$. The expansion underlying Lemma 2.1 gives, at each inverse power x^{-k}, the coefficient in (2.12)

$$\frac{(-1)^k B_k(a)}{k!} \sum_{r=0}^{\min(q,k-1)} (-1)^r P_{k-1}^{(r)}(0) L^{q-r} \sum_{j=r}^{q} \binom{j}{r} d_j.$$

For $1 \leq k \leq q$ this polynomial in L must vanish. A nonzero Bernoulli polynomial cannot vanish throughout the prescribed open a-interval. Choosing $k = r+1$ for $r = 0, \ldots, q-1$ and using $P_r^{(r)}(0) = r!$ therefore forces $d^{(r)}(1) = 0$. Thus $d(t)$ is a constant multiple of $(1-t)^q$; $d(0) = 1$ fixes the constant. The remaining leading logarithm vanishes precisely when

$$b = \sum_{j=0}^{q} \frac{(-1)^j \binom{q}{j}}{j+1} = \int_0^1 (1-t)^q \, \mathrm{d}t = \frac{1}{q+1}.$$

Conversely these coefficients have the required decay by Lemma 2.1. □

This uniqueness statement is restricted to (2.12). It does not classify arbitrary combinations with rational x-dependent coefficients or arbitrary differential operators.

2.3 Exact shift and averaging relations

The elementary Hurwitz recurrence gives

$$\boxed{\mathcal{C}_q(x, a+1) - \mathcal{C}_q(x,a) = -\frac{\log^q(x/(x+a))}{x+a}.} \tag{2.14}$$

This is an identity of ordinary functions, including $q = 0$. Integrating $\partial_a \zeta(z, x+a) = -z\zeta(1+z, x+a)$ over $0 \leq a \leq 1$ instead gives, for each fixed $x > 0$,

$$\int_0^1 F(z; x, a) \, \mathrm{d}a = 0, \qquad \int_0^1 \mathcal{C}_q(x,a) \, \mathrm{d}a = 0. \tag{2.15}$$

The apparent $z = 0$ singularity is removed before differentiation. Equation (2.15) concerns a finite shift interval; it does not assert an unproved interchange of two unbounded integrals.

3 The Mellin theorem and its analytic proof

3.1 The harmonic coefficient

For $\operatorname{Re} u > 0$, define $Q_q(u)$ by (1.3) near $z = 0$. Set

$$\begin{aligned} H^{(1)}_{u-1} &= \psi(u) + \gamma, \\ H^{(r)}_{u-1} &= \zeta(r) - \zeta(r, u) \qquad (r \geq 2). \end{aligned} \tag{3.1}$$

These continue the usual generalized harmonic numbers. The Gamma Taylor formula gives

$$\frac{\Gamma(u+z)}{\Gamma(u)\Gamma(1+z)} = \exp\left(\sum_{r\geq 1} \frac{(-1)^{r-1}}{r} H^{(r)}_{u-1} z^r\right). \tag{3.2}$$

Consequently

$$\begin{aligned} Q_0 &= 1, & Q_1 &= H^{(1)}, \\ Q_2 &= (H^{(1)})^2 - H^{(2)}, & Q_3 &= (H^{(1)})^3 - 3H^{(1)}H^{(2)} + 2H^{(3)}. \end{aligned} \tag{3.3}$$

At a positive integer m,

$$\frac{\Gamma(m+z)}{\Gamma(m)\Gamma(1+z)} = E_m(z) := \prod_{j=1}^{m-1} \left(1 + \frac{z}{j}\right), \qquad Q_q(m) = q! e^{(m)}_q, \tag{3.4}$$

where $e^{(m)}_q = [z^q]E_m(z)$ and $e^{(m)}_q = 0$ for $q \geq m$. This is an *elementary*, not complete homogeneous, harmonic polynomial. The distinction will reverse in a different direction of the mixed generating function in Section 7.

Theorem 3.1 (Balanced Mellin transform)**.** *For $a > 0$, $q \geq 0$, and $0 < \operatorname{Re} u < q + 1$,*

$$\begin{aligned} \mathcal{J}_q(u, a) &= \Gamma(1-u)\zeta(1-u, a)\, \partial^q_z \frac{\Gamma(u+z)}{\Gamma(1+z)}\bigg|_{z=0} \\ &= \frac{\pi}{\sin \pi u}\, \zeta(1-u, a) Q_q(u). \end{aligned} \tag{3.5}$$

At $u = 1, \ldots, q$, the entire right-hand expression is evaluated by its removable limit. These are values of absolutely convergent integrals.

Proof. First take $\operatorname{Re} p > 0$ and $\operatorname{Re} v > \operatorname{Re} p + 1$. Termwise beta integration, justified by absolute convergence, gives

$$\int_0^\infty x^{p-1} \zeta(v, x+a)\, \mathrm{d}x = \frac{\Gamma(p)\Gamma(v-p)}{\Gamma(v)} \zeta(v-p, a). \tag{3.6}$$

Indeed the integral of $(x+n+a)^{-v}$ is $(n+a)^{p-v} B(p, v-p)$.

We need the adjacent strip, not an unjustified substitution into this integral. Split at $x = 1$ and subtract only the leading tail there. The meromorphic continuation of the left side of (3.6) is

$$\begin{aligned} &\int_0^1 x^{p-1} \zeta(v, x+a)\, \mathrm{d}x \\ &\quad + \int_1^\infty x^{p-1} \left(\zeta(v, x+a) - \frac{x^{1-v}}{v-1}\right) \mathrm{d}x + \frac{1}{(v-1)(v-p-1)}. \end{aligned} \tag{3.7}$$

The subtracted tail converges for $\operatorname{Re} p < \operatorname{Re} v$. In the overlapping strip

$$\operatorname{Re} p > 0, \qquad \operatorname{Re}(v-1) < \operatorname{Re} p < \operatorname{Re} v,$$

the last term in (3.7) is the negative of $\int_0^1 x^{p-v}\,\mathrm{d}x/(v-1)$. Thus the continued expression is the *ordinary* whole-axis difference integral

$$\int_0^\infty x^{p-1}\left(\zeta(v, x+a) - \frac{x^{1-v}}{v-1}\right)\mathrm{d}x = \frac{\Gamma(p)\Gamma(v-p)}{\Gamma(v)}\zeta(v-p, a). \tag{3.8}$$

This cutoff argument is a standard Mellin-continuation principle [6]; it specifies all terms and does not assign a value to the separate divergent integral of a monomial.

Now put $v = 1+z$ and $p = u+z$. For $0 < \operatorname{Re} u < 1$ and z sufficiently near zero, the left side becomes

$$\int_0^\infty x^{u-1}F(z; x, a)\,\mathrm{d}x,$$

and the right side becomes

$$\frac{\Gamma(u+z)\Gamma(1-u)}{\Gamma(1+z)}\zeta(1-u, a). \tag{3.9}$$

Near infinity, Lemma 2.1 is uniform in z. Near zero, choose $|\operatorname{Re} z| < \operatorname{Re} u/2$; the regularized leading term $(x^z-1)/z$ is bounded by a constant times $(1+|\log x|)x^{-|\operatorname{Re} z|}$, and the remaining local Hurwitz family has the same integrable type. Cauchy differentiation or dominated parameter differentiation now gives the first line of (3.5) on $0 < \operatorname{Re} u < 1$.

Corollary 2.2 extends the left side holomorphically to $0 < \operatorname{Re} u < q+1$. The right side is meromorphic there. Analytic continuation makes the two equal and forces every pole in this strip to be removable. Finally, Euler's reflection formula $\Gamma(u)\Gamma(1-u) = \pi/\sin \pi u$ gives the second line. □

Remark 3.2 (Why the differentiated strip is larger). The undifferentiated integral of F generally has only the strip $0 < \operatorname{Re} u < 1$. Differentiating it there is legitimate. The *resulting* function $\mathcal{C}_q$ has a larger convergence strip because derivatives annihilate finite-degree Pochhammer polynomials. One must not differentiate a divergent integral at an arbitrary integer u and call that the proof.

4 All positive-integer moments

Theorem 4.1 (Bernoulli closure). *For $a > 0$ and integers $q \geq m \geq 1$, the coefficient $\kappa_{m,q}$ in (1.4) is finite and nonzero, and*

$$\mathcal{J}_q(m, a) = \kappa_{m,q}B_m(a). \tag{4.1}$$

Equivalently,

$$\kappa_{m,q} = \frac{(-1)^{q+1}q!}{m}\sum_{n=m}^{\infty}\frac{\binom{n-1}{m-1}}{n^{q+1}}. \tag{4.2}$$

The infinite series here is a second representation of a finite zeta combination, not an additional unresolved constant.

Proof. At $u = m$, (3.4) gives $Q_q(m) = 0$. Since $\pi \csc \pi u = (-1)^m/(u-m) + O(u-m)$, Theorem 3.1 gives

$$\mathcal{J}_q(m, a) = \frac{(-1)^{m+1}}{m}B_m(a)Q_q'(m). \tag{4.3}$$

We used the classical value $\zeta(1-m,a) = -B_m(a)/m$. Differentiating the Gamma quotient in u gives

$$Q_q'(m) = q![z^q]E_m(z)(\psi(m+z) - \psi(m)). \tag{4.4}$$

The digamma recurrence writes the last factor as

$$\psi(1+z) - \psi(1) + \sum_{j=1}^{m-1} \left(\frac{1}{j+z} - \frac{1}{j}\right).$$

After multiplication by E_m, the finite rational sum is a polynomial of degree at most $m-1$. It cannot contribute at degree $q \geq m$. The Taylor series

$$\psi(1+z) - \psi(1) = \sum_{r\geq 1} (-1)^{r+1}\zeta(r+1)z^r$$

therefore gives

$$Q_q'(m) = q! \sum_{j=0}^{m-1} e_j^{(m)} (-1)^{q-j+1}\zeta(q-j+1). \tag{4.5}$$

Substitution into (4.3) proves (4.1) with exactly the sign in (1.4).

To obtain (4.2), expand every zeta value into its absolutely convergent series. The coefficient at n is

$$\sum_{j=0}^{m-1} (-1)^j e_j^{(m)} n^j = \prod_{j=1}^{m-1} (1 - n/j) = (-1)^{m-1}\binom{n-1}{m-1}.$$

It vanishes at $n = 1, \ldots, m-1$. The remaining series converges since $q \geq m$, and all its terms are positive. Thus $\kappa_{m,q} \neq 0$. □

4.1 Explicit evaluations and a centered vanishing family

For every $q \geq 1$,

$$\boxed{\int_0^\infty \mathcal{C}_q(x,a)\,\mathrm{d}x = (-1)^{q+1}q!(a-\tfrac{1}{2})\zeta(q+1).} \tag{4.6}$$

In particular,

$$\int_0^\infty \left[\frac{\log^{q+1} x}{q+1} + \sum_{j=0}^{q} (-1)^j \binom{q}{j} \gamma_j(x+\tfrac{1}{2}) \log^{q-j} x\right] \mathrm{d}x = 0 \quad (q \geq 1). \tag{4.7}$$

Each complete bracket is integrated as one function. Its summands need not have separate convergent integrals.

At $q = 1$, $a = 1$, this gives

$$\int_0^\infty \left[-\psi(x+1)\log x - \gamma_1(x+1) + \frac{1}{2}\log^2 x\right] \mathrm{d}x = \frac{\pi^2}{12}. \tag{4.8}$$

At $q = 2$ the first weighted evaluation is

$$\int_0^\infty x\left[-\psi(x+a)\log^2 x - 2\gamma_1(x+a)\log x + \gamma_2(x+a) + \frac{1}{3}\log^3 x\right] \mathrm{d}x$$
$$= (\zeta(3) - \zeta(2))(a^2 - a + \tfrac{1}{6}). \tag{4.9}$$

A compact table of multipliers is

m	q	$\kappa_{m,q}$
1	1	$\zeta(2)$
1	2	$-2\zeta(3)$
1	3	$6\zeta(4)$
2	2	$\zeta(3)-\zeta(2)$
2	3	$3\zeta(3)-3\zeta(4)$
2	4	$12\zeta(5)-12\zeta(4)$
3	3	$\zeta(2)-3\zeta(3)+2\zeta(4)$
3	4	$-4\zeta(3)+12\zeta(4)-8\zeta(5)$
4	4	$-\zeta(2)+6\zeta(3)-11\zeta(4)+6\zeta(5)$

The small size of these expressions is not a low-order accident: the finite product E_m fixes their length for every $q \geq m$.

4.2 Reflection and moment rank

For $0 < a < 1$, Bernoulli reflection gives

$$\int_0^\infty x^{m-1}\big(\mathcal{C}_q(x,a) - (-1)^m \mathcal{C}_q(x,1-a)\big)\,\mathrm{d}x = 0, \qquad q \geq m. \tag{4.10}$$

Thus every odd $m \leq q$ gives a centered vanishing moment at $a = 1/2$. For fixed $q \geq M$, the M moment functions of a, $\mathcal{J}_q(1,a), \ldots, \mathcal{J}_q(M,a)$, are linearly independent polynomials. Indeed they are nonzero scalar multiples of $B_1(a), \ldots, B_M(a)$, whose degrees are distinct. This is an exact functional-rank statement, not a claim about the arithmetic independence of special values.

5 Exceptional moments and polylogarithm derivatives

Theorem 5.1 (The Bernoulli-root moment)**.** *Let $m \geq 1$ and $a > 0$ satisfy $B_m(a) = 0$. Then the ordinary integral*

$$\boxed{\int_0^\infty x^{m-1}\mathcal{C}_{m-1}(x,a)\,\mathrm{d}x = (-1)^{m-1}\zeta^{[1]}(1-m,a)} \tag{5.1}$$

converges absolutely.

Proof. Lemma 2.1 removes the leading x^{-m} term, so the integrand at infinity is $O(x^{-2})$. The integral is holomorphic in u in a neighborhood of $u = m$. Its value follows from Theorem 3.1 by continuation from $\operatorname{Re} u < m$. Here

$$Q_{m-1}(m) = (m-1)!\,e_{m-1}^{(m)} = 1, \qquad \zeta(1-u,a) = -(u-m)\zeta^{[1]}(1-m,a) + O((u-m)^2).$$

Multiplication by $\pi \csc \pi u$ gives (5.1). □

This is a different mechanism from Theorem 4.1. There, $Q_q(m)$ vanishes; here it does not, and the Hurwitz value vanishes instead. Substituting $q = m-1$ into the finite sum (1.4) would introduce $\zeta(1)$. Multiplying that undefined expression by $B_m(a) = 0$ is not a valid limiting argument.

5.1 Centered and algebraic-shift examples

For $m = 1$, the only root of B_1 is $a = 1/2$. The root formula becomes

$$\int_0^\infty \left(\log x - \psi(x + \tfrac{1}{2})\right) \mathrm{d}x = \zeta^{[1]}(0, \tfrac{1}{2}) = -\frac{1}{2}\log 2. \tag{5.2}$$

This basic log-Gamma consequence is classical, not a novelty claim. It is a useful normalization check on the higher-order family.

For $r \geq 1$, both $B_{2r+1}(1)$ and $B_{2r+1}(1/2)$ vanish. Thus

$$\int_0^\infty x^{2r} \mathcal{C}_{2r}(x, 1) \, \mathrm{d}x = \frac{(-1)^r (2r)!}{2(2\pi)^{2r}} \zeta(2r + 1), \tag{5.3}$$

$$\int_0^\infty x^{2r} \mathcal{C}_{2r}(x, \tfrac{1}{2}) \, \mathrm{d}x = (2^{-2r} - 1) \frac{(-1)^r (2r)!}{2(2\pi)^{2r}} \zeta(2r + 1). \tag{5.4}$$

These use the standard zeta functional equation [4] and $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$. At $r = 1$,

$$\int_0^\infty x^2 \mathcal{C}_2(x, 1) \, \mathrm{d}x = -\frac{\zeta(3)}{4\pi^2}, \tag{5.5}$$

$$\int_0^\infty x^2 \mathcal{C}_2(x, \tfrac{1}{2}) \, \mathrm{d}x = \frac{3\zeta(3)}{16\pi^2}. \tag{5.6}$$

At the two positive roots $a_\pm = (1 \pm 1/\sqrt{3})/2$ of B_2,

$$\int_0^\infty x \mathcal{C}_1(x, a_\pm) \, \mathrm{d}x = -\zeta^{[1]}(-1, a_\pm). \tag{5.7}$$

This is an exact reduction to a specified Hurwitz derivative. No further elementary or cyclotomic evaluation at these algebraic shifts is inferred.

5.2 Every logarithmic moment at a Bernoulli root

The extra convergence in Theorem 5.1 persists after insertion of any fixed power of $\log x$. It gives more than a single boundary value.

Corollary 5.2 (All exceptional logarithmic moments)**.** *Under the hypotheses of Theorem 5.1, for every $r \geq 0$,*

$$\begin{aligned} &\int_0^\infty x^{m-1} \log^r x \, \mathcal{C}_{m-1}(x, a) \, \mathrm{d}x \\ &\quad = (-1)^{m-1} r! [t^r] \left\{ \frac{\pi t}{\sin \pi t} Q_{m-1}(m + t) \sum_{j=0}^{r} \frac{(-1)^j \zeta^{[j+1]}(1 - m, a)}{(j + 1)!} t^j \right\}. \end{aligned} \tag{5.8}$$

In particular, with

$$\alpha_m = (m - 1)! \sum_{j=0}^{m-2} (-1)^{m-j} e_j^{(m)} \zeta(m - j) - H_{m-1}, \qquad \alpha_1 = 0, \tag{5.9}$$

the first logarithmic moment is

$$\int_0^\infty x^{m-1} \log x \, \mathcal{C}_{m-1}(x, a) \, \mathrm{d}x = (-1)^{m-1} \left\{ \alpha_m \zeta^{[1]}(1 - m, a) - \frac{1}{2} \zeta^{[2]}(1 - m, a) \right\}. \tag{5.10}$$

Proof. The root condition makes $x^{m-1}\mathcal{C}_{m-1}(x,a) = O(x^{-2})$. Every displayed logarithmic integral therefore converges, and equals the corresponding u-derivative of the Mellin transform at $u = m$. Expand (3.5) there, using $\pi \csc \pi(m+t) = (-1)^m \pi / \sin \pi t$ and the vanishing Hurwitz constant term. This proves (5.8). Since $Q_{m-1}(m) = 1$ and $\pi t / \sin \pi t$ has zero linear coefficient, the first derivative gives (5.10) with $\alpha_m = Q'_{m-1}(m)$. To compute it explicitly, use (4.4) at degree $m-1$. The finite rational recurrence term now contributes $-H_{m-1}$, whereas the digamma Taylor series contributes the finite zeta sum in (5.9). □

Two closed examples are

$$\int_0^\infty \log x \big(\log x - \psi(x + \tfrac{1}{2})\big)\, \mathrm{d}x = \frac{1}{4}\log^2 2 + \frac{1}{2}\log 2 \log(2\pi) \tag{5.11}$$

and

$$\begin{aligned}\int_0^\infty & x^2 \log x\, \mathcal{C}_2(x, \tfrac{1}{2})\, \mathrm{d}x \\ &= \frac{3}{16\pi^2}\left[\zeta^{[1]}(3) + \zeta(3)\left(3\zeta(2) - 2\zeta(3) - \gamma - \log(2\pi) + \frac{\log 2}{3}\right)\right].\end{aligned} \tag{5.12}$$

For the second formula, $\alpha_3 = 3\zeta(2) - 2\zeta(3) - 3/2$. Differentiating the Riemann functional equation at -2 yields

$$\zeta^{[2]}(-2) = 2\zeta^{[1]}(-2)\left(\log(2\pi) - \frac{3}{2} + \gamma - \frac{\zeta^{[1]}(3)}{\zeta(3)}\right).$$

The half-shift duplication formula then gives (5.12). These formulas display explicitly how a logarithmic moment introduces a positive-order zeta derivative.

5.3 An explicit polylogarithmic form

For $0 < a < 1$, define

$$\mathcal{P}_\nu(a) = e^{-\mathrm{i}\pi\nu/2}\,\mathrm{Li}_\nu(e^{2\pi\mathrm{i}a}) + e^{\mathrm{i}\pi\nu/2}\,\mathrm{Li}_\nu(e^{-2\pi\mathrm{i}a}). \tag{5.13}$$

For $\mathrm{Re}\,\nu > 1$, all series and their finitely many order derivatives converge absolutely. Hurwitz's Fourier formula is

$$\zeta(1-\nu, a) = \frac{\Gamma(\nu)}{(2\pi)^\nu}\mathcal{P}_\nu(a) \quad [1, 25.11.9]. \tag{5.14}$$

It uses the standard polylogarithm convention $\mathrm{Li}_\nu(z) = \sum_{n\ge1} z^n/n^\nu$ in its initial disc and its boundary continuation [3].

Corollary 5.3 (Polylogarithmic root identity)**.** *If $m \ge 2$, $0 < a < 1$, and $B_m(a) = 0$, then*

$$\begin{aligned}\int_0^\infty x^{m-1}\mathcal{C}_{m-1}(x,a)\,\mathrm{d}x = \frac{(-1)^m(m-1)!}{(2\pi)^m}\Big\{ & e^{-\mathrm{i}\pi m/2}\left[\mathrm{Li}_m^{[1]}(e^{2\pi\mathrm{i}a}) - \frac{\mathrm{i}\pi}{2}\,\mathrm{Li}_m(e^{2\pi\mathrm{i}a})\right] \\ & + e^{\mathrm{i}\pi m/2}\left[\mathrm{Li}_m^{[1]}(e^{-2\pi\mathrm{i}a}) + \frac{\mathrm{i}\pi}{2}\,\mathrm{Li}_m(e^{-2\pi\mathrm{i}a})\right]\Big\}.\end{aligned} \tag{5.15}$$

Proof. At the root, $\mathcal{P}_m(a) = 0$ by (5.14). Differentiating that formula in ν therefore eliminates the derivative of its Gamma prefactor and gives

$$-\zeta^{[1]}(1-m, a) = \frac{(m-1)!}{(2\pi)^m}\mathcal{P}_m^{[1]}(a).$$

Use Theorem 5.1 and differentiate (5.13). The two terms in braces are conjugates for real a; the displayed value is real. □

Away from the Bernoulli-root condition, the same differentiation retains an additional $(\psi(m) - \log(2\pi))\mathcal{P}_m(a)$ term. Dropping that term without its vanishing hypothesis would give a wrong identity. For ordinary moments with $q \geq m \geq 2$, one may instead substitute $B_m(a) = -m!\mathcal{P}_m(a)/(2\pi)^m$ directly into (4.1). This gives finite polylogarithm coordinates at every rational shift without asserting their arithmetic independence.

6 Finite shift filters: a complete moment criterion

Let $a_1, \ldots, a_J > 0$ and $c_1, \ldots, c_J \in \mathbb{C}$. Write

$$\mathcal{L}f = \sum_{j=1}^{J} c_j f(a_j), \qquad Z_{\mathcal{L}}(s) = \sum_{j=1}^{J} c_j \zeta(s, a_j), \qquad \mathcal{C}_q^{\mathcal{L}}(x) = \sum_{j=1}^{J} c_j \mathcal{C}_q(x, a_j). \tag{6.1}$$

The logarithmic term in $\mathcal{C}_q^{\mathcal{L}}$ vanishes only when $\sum c_j = 0$; until that condition is imposed it is retained.

Theorem 6.1 (Necessary and sufficient cancellation)**.** *Let $m \geq 1$, $q \geq 0$. If $q \geq m$, the moment of $\mathcal{C}_q^{\mathcal{L}}$ always converges absolutely and equals $\kappa_{m,q}\mathcal{L}B_m$. If $q < m$, the ordinary improper integral*

$$\int_0^\infty x^{m-1}\mathcal{C}_q^{\mathcal{L}}(x)\,\mathrm{d}x \tag{6.2}$$

converges if and only if

$$\mathcal{L}B_k = 0 \qquad (k = q+1, \ldots, m). \tag{6.3}$$

When it converges, it is absolutely convergent and

$$\boxed{\int_0^\infty x^{m-1}\mathcal{C}_q^{\mathcal{L}}(x)\,\mathrm{d}x = (-1)^{m-1}q!e_q^{(m)}Z_{\mathcal{L}}^{[1]}(1-m).} \tag{6.4}$$

Proof. For $k \geq q+1$, $P_{k-1}^{(q)}(0) > 0$: its polynomial has strictly positive coefficients in all degrees from 0 to $k-1$. Hence the coefficients of x^{-k} in Lemma 2.1 vanish for the filter exactly when $\mathcal{L}B_k = 0$. If the first surviving power has $k \leq m$, the weighted integrand has a nonzero leading constant multiple of x^{m-1-k}, whose integral diverges (logarithmically for $k = m$). The lower powers of x cannot cancel this divergence. This remains true for complex weights because at least one real component has the nonzero leading coefficient. If all powers through $k = m$ vanish, the integrand is $O(x^{-2})$. The zero endpoint is integrable as before. This proves necessity, sufficiency, and absolute convergence.

The filtered Mellin theorem initially holds on $0 < \operatorname{Re} u < q+1$ by finite summation. Under (6.3) its left side extends holomorphically through $u = m$. Since $Z_{\mathcal{L}}(1-m) = -\mathcal{L}B_m/m = 0$ and $Q_q(m) = q!e_q^{(m)}$, its removable limit is (6.4). The case $q \geq m$ is Theorem 4.1 summed over the shifts. □

6.1 Polynomial annihilators and minimal support

Suppose $\mathcal{L}$ annihilates every polynomial of degree at most m. Equivalently,

$$\sum_{j=1}^{J} c_j a_j^k = 0 \qquad (k = 0, \dots, m). \tag{6.5}$$

It then annihilates $B_0, \dots, B_m$. Every balanced moment of order m exists and

$$\int_0^\infty x^{m-1} \mathcal{C}_q^{\mathcal{L}}(x)\, \mathrm{d}x = \begin{cases} (-1)^{m-1} q!\, e_q^{(m)}\, Z_{\mathcal{L}}^{[1]}(1-m), & 0 \le q < m, \\ 0, & q \ge m. \end{cases} \tag{6.6}$$

For distinct prescribed nodes, a nonzero filter satisfying (6.5) requires at least $m+2$ nodes. Indeed, the Vandermonde matrix on any $m+1$ distinct nodes is invertible. With $m+2$ nodes the nullspace is one-dimensional, so the normalized annihilator is unique. This is a statement about full polynomial annihilation, not about the weaker, potentially shorter list in (6.3).

For equally spaced nodes one may take the forward difference

$$\Delta_h^{m+1} f(a) = \sum_{j=0}^{m+1} (-1)^{m+1-j} \binom{m+1}{j} f(a + jh), \qquad a, h > 0. \tag{6.7}$$

Thus an entire infinite sequence of balanced moments vanishes after one fixed finite difference.

6.2 Concrete filtered identities

Let

$$\mathcal{L} f = f(\tfrac{1}{2}) - 2f(1) + f(\tfrac{3}{2}). \tag{6.8}$$

It kills constants and linear polynomials. At $m = 1$, $q = 0$,

$$\int_0^\infty \left[-\psi(x + \tfrac{1}{2}) + 2\psi(x+1) - \psi(x + \tfrac{3}{2}) \right] \mathrm{d}x = \log \frac{\pi}{2}. \tag{6.9}$$

At every $q \ge 1$, the same filter applied to $\mathcal{C}_q$ has integral zero. The raw Stieltjes functions require the additional logarithmic moment conversion developed next.

For the third difference with step $1/2$ based at $1/2$, put

$$\mathcal{L}_3 f = -f(\tfrac{1}{2}) + 3f(1) - 3f(\tfrac{3}{2}) + f(2).$$

The recurrence and duplication formulas give

$$Z_{\mathcal{L}_3}(s) = 4(2 - 2^s)\zeta(s) + 3 \cdot 2^s - 1.$$

Its $m = 2$ moments at $q = 0$ and $q = 1$ are both

$$\int_0^\infty x \mathcal{C}_q^{\mathcal{L}_3}(x)\, \mathrm{d}x = -6\zeta^{[1]}(-1) - \frac{5}{3}\log 2 \qquad (q = 0, 1), \tag{6.10}$$

and every $q \ge 2$ moment is zero. The identical first two values arise from $0! e_0^{(2)} = 1! e_1^{(2)} = 1$, not from a presumed equality of the two functions.

7 All mixed logarithmic moments of filtered constants

The finite-filter theorem concerns the balanced combinations. A stronger local generating function recovers the separate filtered Stieltjes constants with arbitrary powers of $\log x$.

Theorem 7.1 (Complete mixed-moment formula)**.** *Let $m \geq 1$ and let $\mathcal{L}$ satisfy (6.5). For all integers $n, r \geq 0$, the ordinary integral*

$$I_{n,r}^{(m)}(\mathcal{L}) = \int_0^\infty x^{m-1} \log^r x \sum_j c_j \gamma_n(x + a_j)\, \mathrm{d}x \tag{7.1}$$

is absolutely convergent. Put $w = \varepsilon - \eta$ and define the holomorphic germ

$$\begin{aligned} \mathcal{K}_m(\varepsilon, \eta) &= \frac{\Gamma(1+\eta)\Gamma(1+w)}{\Gamma(1+\varepsilon)} \prod_{j=1}^{m-1} \frac{1+\eta/j}{1-w/j}, \\ M_m(\varepsilon, \eta) &= (-1)^{m-1} \mathcal{K}_m(\varepsilon, \eta) \frac{Z_{\mathcal{L}}(1-m+w)}{w}. \end{aligned} \tag{7.2}$$

The quotient in the second line is removable because $Z_{\mathcal{L}}(1-m) = 0$. Then

$$\boxed{I_{n,r}^{(m)}(\mathcal{L}) = (-1)^n n! r! [\varepsilon^n \eta^r] M_m(\varepsilon, \eta).} \tag{7.3}$$

Every fixed coefficient requires only $Z_{\mathcal{L}}^{[k]}(1-m)$ for $1 \leq k \leq n+r+1$, ordinary zeta values, and finite generalized harmonic numbers.

Proof. Because the filter kills every polynomial through degree m, the first nonzero possible term in the large-x expansion of $\sum c_j \zeta(1+\varepsilon, x+a_j)$ has index $k = m+1$. Its leading integral term also cancels since $\sum c_j = 0$. On a small compact neighborhood of $(\varepsilon, \eta) = (0,0)$, multiplication by $x^{m+\eta-1}$ therefore gives an integrable bound of type $x^{-2+\delta}$, for some $0 < \delta < 1$. Every parameter derivative adds only a fixed power of $\log x$. At zero, all shifts are positive and $x^{m+\eta-1}$ supplies the required integrability. The filtered integral is consequently jointly holomorphic near the origin.

More explicitly, the same endpoint estimates show that the filtered integral is jointly holomorphic on the connected tube $\mathrm{Re}(m+\eta) > 0$, $\mathrm{Re}(\varepsilon - \eta) > -1$. The pole at $\varepsilon = 0$ cancels because $\sum c_j = 0$. This tube contains the initial absolute beta domain $\mathrm{Re}\,\varepsilon > m + \mathrm{Re}\,\eta$. Summing (3.6) there and continuing on this tube gives

$$\int_0^\infty x^{m+\eta-1} \sum_j c_j \zeta(1+\varepsilon, x+a_j)\, \mathrm{d}x = \frac{\Gamma(m+\eta)\Gamma(1-m+w)}{\Gamma(1+\varepsilon)} Z_{\mathcal{L}}(1-m+w). \tag{7.4}$$

The possible Gamma poles in this tube occur at $w = 0, 1, \ldots, m-1$. At each such point the corresponding $Z_{\mathcal{L}}(1-m+w)$ vanishes, since it is a multiple of one of the annihilated Bernoulli polynomials $B_m, \ldots, B_1$. Thus the complete right side is holomorphic as well, and the identity theorem applies. Equivalently, finite Euler–Maclaurin subtraction makes the same continuation explicit. No unspecified finite-part constant is introduced.

The recurrence for Gamma factors gives

$$\Gamma(m+\eta) = (m-1)!\Gamma(1+\eta) \prod_{j=1}^{m-1} (1+\eta/j),$$

$$\Gamma(1-m+w)=\frac{(-1)^{m-1}\Gamma(1+w)}{(m-1)!w\prod_{j=1}^{m-1}(1-w/j)}.$$

Thus the right side of (7.4) is precisely M_m. The coefficient of ε^n in the filtered Hurwitz family is $(-1)^n\sum c_j\gamma_n(x+a_j)/n!$, and the coefficient of η^r in x^η is $\log^r x/r!$. Extracting coefficients proves (7.3) and ordinary convergence of all its entries. □

7.1 A finite harmonic and zeta coefficient compiler

The logarithm of the elementary germ in (7.2) is

$$\log\mathcal{K}_m(\varepsilon,\eta)=\sum_{k\geq 2}\frac{(-1)^k\zeta(k)}{k}(\eta^k+w^k-\varepsilon^k)$$
$$+\sum_{k\geq 1}\frac{H_{m-1}^{(k)}}{k}((-1)^{k-1}\eta^k+w^k). \tag{7.5}$$

The Euler constant cancels because $\eta+w-\varepsilon=0$. For a requested total degree $d=n+r$, truncate this logarithm through degree d, exponentiate formally through that degree, and multiply by

$$\frac{Z_{\mathcal{L}}(1-m+w)}{w}=\sum_{j=0}^{d}\frac{Z_{\mathcal{L}}^{[j+1]}(1-m)}{(j+1)!}w^j+O(w^{d+1}). \tag{7.6}$$

No infinite coefficient operation is needed for a fixed identity. The remaining Hurwitz derivatives are named analytic coordinates; this algorithm alone does not prove they reduce to fewer constants.

The two simplest directions through this germ are

$$\mathcal{K}_m(\varepsilon,0)=\prod_{j=1}^{m-1}(1-\varepsilon/j)^{-1}, \tag{7.7}$$

$$\mathcal{K}_m(\varepsilon,\varepsilon)=\prod_{j=1}^{m-1}(1+\varepsilon/j)=E_m(\varepsilon). \tag{7.8}$$

Thus raw order differentiation sees complete homogeneous harmonic coefficients, while balanced diagonal differentiation sees elementary ones. This explains both the finite degree and the infinite vanishing family in (6.6).

Corollary 7.2 (Raw moments without an extra logarithm)**.** *Let $h_j^{(m)}$ be defined by $\prod_{k=1}^{m-1}(1-z/k)^{-1}=\sum_{j\geq 0}h_j^{(m)}z^j$. Under the hypotheses of Theorem 7.1,*

$$I_{n,0}^{(m)}(\mathcal{L})=(-1)^{n+m-1}n!\sum_{j=0}^{n}h_{n-j}^{(m)}\frac{Z_{\mathcal{L}}^{[j+1]}(1-m)}{(j+1)!}. \tag{7.9}$$

In particular,

$$\int_0^\infty\sum_j c_j\gamma_n(x+a_j)\,\mathrm{d}x=\frac{(-1)^n}{n+1}Z_{\mathcal{L}}^{[n+1]}(0) \tag{7.10}$$

when $\sum c_j=\sum c_ja_j=0$.

Proof. Use (7.7) and (7.6). For $m=1$ the finite product is empty, so only $h_0^{(1)}=1$ survives. □

7.2 A closed first-Stieltjes integral

For the half-step filter (6.8),

$$Z_{\mathcal{L}}(s) = 2(2^s - 2)\zeta(s) - 2^s. \tag{7.11}$$

The $n = 1$ case of (7.10) is therefore

$$\begin{aligned}\int_0^\infty [\gamma_1(x+\tfrac{1}{2}) - 2\gamma_1(x+1) + \gamma_1(x+\tfrac{3}{2})]\,\mathrm{d}x \\ = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{24} + \frac{3}{2}\log^2 2 - \frac{1}{2}\log^2 \pi.\end{aligned} \tag{7.12}$$

Here $\gamma_1 = \gamma_1(1)$. To check the constants, differentiating (7.11) twice gives

$$-\tfrac{1}{2}Z_{\mathcal{L}}^{[2]}(0) = \zeta^{[2]}(0) - 2\log 2\,\zeta^{[1]}(0) + \log^2 2.$$

Expansion of the standard Riemann functional equation at zero gives

$$\zeta^{[2]}(0) = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{24} - \frac{1}{2}\log^2(2\pi), \qquad \zeta^{[1]}(0) = -\tfrac{1}{2}\log(2\pi).$$

Substitution proves (7.12). Every higher n similarly has the exact generating description (7.10)–(7.11); expansion of the same functional equation converts it to finitely many ordinary Stieltjes constants and elementary constants. This is an all-order reduction mechanism, not just the displayed first case.

For a genuinely mixed example with $m = 1$, formula (7.5) starts with $\log \mathcal{K}_1 = -\zeta(2)\eta(\varepsilon - \eta) + O((|\varepsilon| + |\eta|)^3)$. Thus any filter killing constants and linear polynomials satisfies

$$\int_0^\infty \log x \sum_j c_j \gamma_1(x + a_j)\,\mathrm{d}x = \frac{1}{3}Z_{\mathcal{L}}^{[3]}(0) + \zeta(2)Z_{\mathcal{L}}^{[1]}(0). \tag{7.13}$$

The $\zeta(2)$ correction comes from the Gamma quotient. Omitting it would confuse a raw moment with a balanced one.

8 Shift derivatives and anchored primitives

Theorem 8.1 (Every shift derivative of the ordinary moments)**.** *For $q \geq m \geq 1$, $a > 0$, and $p \geq 0$, differentiation in a under the integral is legitimate and*

$$\int_0^\infty x^{m-1}\partial_a^p \mathcal{C}_q(x,a)\,\mathrm{d}x = \begin{cases} \kappa_{m,q}\dfrac{m!}{(m-p)!}B_{m-p}(a), & 0 \leq p \leq m, \\ 0, & p > m. \end{cases} \tag{8.1}$$

For $p \geq 1$ the integrand has the explicit form

$$\partial_a^p \mathcal{C}_q(x,a) = (-1)^p\,\partial_z^q\,[x^z(1+z)_p\zeta(p+1+z, x+a)]\big|_{z=0}. \tag{8.2}$$

Proof. The identity $\partial_a^p\zeta(s,a) = (-1)^p(s)_p\zeta(s+p,a)$ gives (8.2). The Euler–Maclaurin expansion used above is also normally uniform for a in a small complex neighborhood of any compact positive interval. Cauchy differentiation therefore bounds every fixed a-derivative by the same integrable endpoint type, possibly after taking extra expansion terms. One may differentiate (4.1) under the integral. Finally $B_m^{(p)}(a) = m!B_{m-p}(a)/(m-p)!$ for $p \leq m$ and vanishes for $p > m$. □

For example,

$$\int_0^\infty \left[-\psi'(x+a)\log x - \gamma_1'(x+a)\right] \mathrm{d}x = \zeta(2),$$
$$\int_0^\infty \left[-\psi''(x+a)\log x - \gamma_1''(x+a)\right] \mathrm{d}x = 0. \tag{8.3}$$

At the level of individual Stieltjes functions,

$$\gamma_n^{(p)}(a) = (-1)^{n+p} n! [z^n](1+z)_p \zeta(p+1+z, a), \qquad p \geq 1. \tag{8.4}$$

This is a finite sum of Hurwitz order derivatives with coefficients from a harmonic polynomial. At $n = 0$ it reduces to the usual polygamma relation. For $n > 0$ the order derivatives must not be silently replaced by polygamma values alone.

Theorem 8.2 (Anchored primitive and its moments)**.** *For $a, b > 0$, set*

$$\mathcal{A}_q(x; a, b) = -\frac{1}{q+1}\, \partial_z^{q+1} \left\{ x^z \left[\zeta(z, x+a) - \zeta(z, x+b)\right] \right\}\Big|_{z=0}. \tag{8.5}$$

Then $\mathcal{A}_q(x; b, b) = 0$ and $\partial_a \mathcal{A}_q = \mathcal{C}_q(x, a)$. Moreover, for $q \geq m \geq 1$,

$$\int_0^\infty x^{m-1} \mathcal{A}_q(x; a, b)\, \mathrm{d}x = \frac{\kappa_{m,q}}{m+1} \left(B_{m+1}(a) - B_{m+1}(b)\right). \tag{8.6}$$

All integrals in this statement are ordinary convergent integrals.

Proof. Differentiate (8.5) in a and use $\partial_a \zeta(z, x+a) = -z\zeta(1+z, x+a)$. By (2.4), the coefficient of z^{q+1} in $z x^z \zeta(1+z, x+a)$ is $\mathcal{C}_q(x, a)/q!$. The derivative of order $q+1$, divided by $q+1$, is exactly $\mathcal{C}_q$. The anchoring is immediate. Thus $\mathcal{A}_q(x; a, b) = \int_b^a \mathcal{C}_q(x, t)\, \mathrm{d}t$. Uniform absolute convergence on the compact shift path permits Fubini and (4.1); integration of B_m proves the moment. □

For direct use, the primitive is the finite expression

$$\mathcal{A}_q(x; a, b) = -\frac{1}{q+1} \sum_{j=0}^{q+1} \binom{q+1}{j} \log^{q+1-j} x \left[\zeta^{[j]}(0, x+a) - \zeta^{[j]}(0, x+b)\right]. \tag{8.7}$$

The terms $j = 0, 1$ are elementary and log-Gamma coordinates. For example,

$$\mathcal{A}_0(x; a, b) = (a-b)\log x - \log \frac{\Gamma(x+a)}{\Gamma(x+b)}. \tag{8.8}$$

Its derivative is the ordinary digamma kernel (2.5). Formula (8.7) specifies the integration constant rather than hiding it in an unspecified antiderivative.

9 Source audit, normalization, and computational evidence

9.1 Scope of the source comparison

The source inspection covered the canonical manuscript README, the harmonic-generating and further-research sections [8, 9], the Mellin continuation [10], selected related chapter text, and

connector searches for overlapping Stieltjes, Mellin, Bernoulli, and moment developments. The repository was changing during retrieval. References therefore record the actual snapshots or blob identifiers used rather than pretending that all files came from one frozen checkout.

The incoming directory listing was inspected. It included archives named *Bernoulli Tails and Spectral Faces*, *Higher Spectral Defects*, *Vertical Lattice Jets*, and other continuations. Their binary archive interiors were not available in the working environment for this session. Consequently we do not claim an exhaustive comparison against those interiors or every preserved report. No theorem in this article depends on an uninspected incoming claim.

The manuscript README correctly separates experimental evidence from accepted proofs and retains S_6 and revised S_8 as conjectural [7]. The harmonic-generating source correctly warns that its finite-part constant does not determine all mixed harmonic Stieltjes coefficients. We found no mathematical error in those specific inspected statements.

9.2 An actual editorial correction

In the inspected README blob `dc0fa26d0e5c2ede56df37c5640c7319681dfd25`, the paragraph beginning with the complete 1,032-input first-letter X family, top-sector rank 1,008, and kernel dimension 24 appears twice consecutively. The proposed correction is to retain one copy, checking the current file before applying a patch. This is an editorial duplication, not a contradiction of the mathematical rank statement. The package contains an integration note; no repository file was changed.

9.3 Mathematical traps prevented by the present proofs

There are three distinct cautions, none attributed to an existing source without an actual occurrence.

First, the separate summands of (1.1) generally diverge when integrated on $(0,\infty)$. The theorem concerns the combined function. Second, the $q = m-1$ boundary must not be evaluated by substituting into (1.4): its last zeta value is $\zeta(1)$, and the Bernoulli-root limit instead gives a zeta derivative. Third, complete and elementary harmonic coefficients arise in *different* directions of (7.2); interchanging them destroys both the finite-degree cancellation and the raw logarithmic corrections.

These warnings have explicit checks. At $q=0, m=1, a=1$, the tail is $-1/(2x)$, so the proposed ordinary moment does not exist. At $q=0, m=1, a=1/2$, the tail cancels and the correct value is $-\log 2/2$, not zero. At $m=1$, the mixed raw moment (7.13) contains the nonzero Gamma correction $\zeta(2)Z_{\mathcal{L}}^{[1]}(0)$.

9.4 Independent exact checks

The included `verify_exact.py` performs exact rational and symbolic comparisons. The independent constructions are integer multiplication of Pochhammer polynomials versus symbolic products; finite harmonic products versus a logarithmic Bell recurrence; re-expansion of the Euler–Maclaurin terms versus Bernoulli polynomials; exact linear systems for uniqueness; polynomial binomial identities for the zeta multipliers; Vandermonde and finite-difference tests; and elementary versus complete harmonic expansions of the two mixed-germ restrictions. Two corruption controls deliberately reject wrong coefficient patterns.

The delivered run passes **823 of 823 exact assertions**. These finite checks verify the reported

algebraic instances and the implementation; they do not constitute proof-assistant formalization of the all-index analytic statements. Those statements are supported by the proofs in the preceding sections.

9.5 Stable numerical evaluation

Directly subtracting several large logarithmic Stieltjes terms at a large x is numerically wasteful. The verification code instead uses an Euler–Maclaurin formula before taking the balanced derivative. Let $A = x + a + N$ and $L = -\log(1 + (a+N)/x)$. With R correction terms,

$$\begin{aligned}\mathcal{C}_q(x,a) \;\approx\; &\sum_{n=0}^{N-1} \frac{[-\log(1+(a+n)/x)]^q}{x+a+n} + \frac{L^{q+1}}{q+1} + \frac{L^q}{2A} \\ &+ \sum_{r=1}^{R} \frac{B_{2r}}{(2r)!A^{2r}} \sum_{j=0}^{\min(q,2r-1)} \binom{q}{j} P_{2r-1}^{(j)}(0) L^{q-j}. \end{aligned} \tag{9.1}$$

The approximation sign is deliberate: the implemented remainder is truncated, not interval-enclosed. The analytic identity underlying this approximation has the usual Euler–Maclaurin remainder.

The integral from zero to a finite cutoff T is evaluated by tanh–sinh quadrature. On $(0,1)$ the substitution $x = t^4$ improves resolution of logarithmic endpoints. The remaining tail is integrated termwise from a finite asymptotic expansion (2.9). For a term $x^{-k}\log^r x$, its exact tail moment is

$$\int_T^\infty x^{u-k-1}\log^r x\,\mathrm{d}x = T^{u-k}\sum_{j=0}^{r}\binom{r}{j}\frac{j!\log^{r-j}T}{(k-u)^{j+1}}, \qquad \operatorname{Re} u < k. \tag{9.2}$$

Filtered coefficients are combined *before* this step. Nonintegrable powers are discarded only when their cancellation has already been established analytically; floating-point smallness is not used as a convergence theorem.

The main run uses 55 working decimal digits, $N = 56$, $R = 40$, $T = 64$, and 64 tail powers. Two selected identities are repeated with 10 additional digits and the enlarged parameters $(N, R, T, K) = (72, 50, 80, 80)$. The checks compare the integral side with independently evaluated Mellin, Bernoulli, zeta-derivative, and mixed-germ right sides. Pointwise comparisons with direct Stieltjes evaluations, shift recurrences, and beta integrals provide additional controls. The main run passes **34 of 34 diagnostics**, with maximum scaled residual below 1.19×10^{-53} at the stated settings. Separate scripts add polylogarithm-root, primitive, averaging, closed-constant, and exceptional logarithmic-moment checks. Exact per-case counts and residuals are recorded in `results/numeric_checks.json` and `results/special_checks.json`, together with `results/log_moment_checks.json`. These are numerical diagnostics, not rigorous accuracy certificates.

Direct near-integer order differentiation of the polylogarithm routine also failed the demanded residual tolerance in three exploratory checks. A 12th-order centered stencil at 85 working digits, independently repeated at steps 10^{-5} and 10^{-6}, removes that numerical discrepancy. The superseded diagnostic is preserved in `results/exploratory_polylog_derivative_checks.json`; it is not counted as a successful verification. The proof of the polylogarithmic identity does not depend on either numerical route.

An initial low-resolution quadrature without the endpoint treatment did not meet the accuracy target. It was not counted as verification. This illustrates why high working precision alone does not certify an improper integral. The delivered script records its method and parameters explicitly.

10 Further research questions

1. Minimal filters for a prescribed collection of moments. Theorem 6.1 gives the exact Bernoulli constraints for one pair (m, q). For a finite set of pairs, determine the minimal support and a canonical normalized solution for positive prescribed nodes. The full polynomial annihilator is sufficient but can impose more conditions than necessary. An exact classification of these smaller Bernoulli evaluation matrices would sharpen the present finite-filter construction without invoking arithmetic independence.

2. Contour transport for complex shifts. The local analytic arguments extend to $\operatorname{Re} a > 0$. Crossing a negative real shift instead moves a Hurwitz singularity through the positive integration axis. Develop contour-deformed analogues of the moment formulas, with explicit contributions from residues and logarithmic cuts. The objective is a branch-consistent identity, not continuation of an unchanged real-axis integral across its singularities.

3. Function-level polylogarithmic identities. The present polylogarithmic evaluations come from the classical Hurwitz functional equation. Seek independent functional relations among $\mathcal{P}_\nu(a)$ and its order derivatives that cause entire families of filtered moments to reduce to fewer coordinates. The right target is an identity in parameters, not merely a small numerical residual at a rational shift.

4. A colored or Lerch analogue of unique balancing. Replace Hurwitz zeta by a Lerch family with a nontrivial color. Its order pole and large-shift expansion change, so the correct balancing need not be (1.1). Classify the homogeneous combinations that cancel the first q tails and determine whether their Mellin moments reduce to Apostol–Bernoulli or other finite polynomial coordinates. Rational colors should be compared with the finite character filters already used in ProveIt.

5. Products of balanced kernels. The scalar Mellin transform is completely explicit. Products $\mathcal{C}_p(x, a)\mathcal{C}_q(x, b)$ lead to Mellin convolutions and potentially coupled multiple-zeta or multiple-Lerch coordinates. Determine a first exact bilinear formula, including an honest function-class statement. A product of two individually reduced transforms does not by itself prove a reduction of the product integral.

6. Relation to genuinely harmonic Stieltjes data. The unresolved repository coefficients $R_{r,k}(a)$ belong to a different family. Investigate whether the translation-variable balancing used here can be combined with its moving-pole subtraction to produce ordinary mixed moments. An answer should retain both regulators and should not identify these coefficients with $\gamma_k(a)$ by notation alone.

7. Higher anchored primitives and Barnes coordinates. Iterating Theorem 8.2 introduces Hurwitz derivatives at successively lower integers. Determine compact anchored formulas using Barnes multiple Gamma functions, and compare normalizations with the raw mixed-moment theorem. Constants at each anchor should be explicit rather than absorbed into unrelated integration constants.

8. Certified computation and formal analytic interfaces. Turn the stable evaluator (9.1) into an outward-rounded algorithm with explicit remainder bounds. Separately, formalize the cutoff

Mellin continuation, locally normal parameter differentiation, and finite-filter convergence criterion. These are the reusable analytic steps; checking additional finite tables alone would not formalize the all-order theorems.

11 Conclusion

The balanced generator $x^z\zeta(1+z, x+a) - 1/z$ removes the order pole and arranges the translation asymptotics into finite-degree Pochhammer polynomials. This converts differentiation in z into exact tail cancellation. Its Mellin transform then exposes two separate resonances: a zero of a harmonic Gamma coefficient gives Bernoulli-polynomial moments, whereas a zero of the Bernoulli polynomial gives an additional zeta-derivative moment. Finite filters make the necessary cancellation conditions explicit and yield a complete mixed logarithmic calculus.

The resulting identities are ordinary convergent integrals with specified constants, not unnormalized finite parts. Their usefulness for ProveIt lies in this combination of exact formulas, parameter proofs, and a reproducible coefficient mechanism. They are offered with a bounded source audit and a clear separation between analytic proof, exact finite verification, numerical diagnostics, and questions that remain unresolved.

A Reproduction and integration map

The package includes the complete source and compiled PDF in `article/`, the mathematical routines and four verification scripts in `code/`, and machine-readable evidence in `results/`. The following commands run from the package root:

```
python code/verify_exact.py
python code/verify_numeric.py
python code/verify_special.py
python code/verify_log_moments.py
python build.py
```

The first script requires SymPy and mpmath; the numerical script uses mpmath. The build requires pdfLaTeX and the packages declared in the preamble. The included `requirements.txt` specifies the versions used for this delivery. The program does not require network access.

Result	Principal proof dependency
Unique balanced kernel	Uniform Hurwitz expansion; finite polynomial roots
Mellin theorem	Beta integral; explicit cutoff continuation; normal convergence
Integer moments	Gamma reflection; finite elementary harmonic coefficients
Exceptional moments	Bernoulli zero; removable Mellin singularity
Finite-filter criterion	Nonzero Pochhammer coefficients; leading-tail test
Mixed raw moments	Filtered beta transform; Gamma recurrence; joint holomorphy
Shift calculus	Hurwitz argument derivatives; compact normal convergence

Suggested placement is a new continuation under the Stieltjes-correlation Mellin material, with cross-references to the harmonic Gamma-coefficient calculus. The report is self-contained and does not require promoting any pending source theorem. Integration should retain its original normalization, in particular the signs in (1.1), the factor $q!$ in (1.3), and the distinction between (4.1) and (5.1). A short integration fragment and source-status record are included. No automatic patch or remote write is performed by the build or verification scripts.

References

[1] NIST Digital Library of Mathematical Functions, *§25.11, Hurwitz Zeta Function*; especially equations 25.11.9, 25.11.12–14, 25.11.17–18, and the Euler–Maclaurin and asymptotic subsections. Accessed 11 October 2026. https://dlmf.nist.gov/25.11.

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[7] ProveIt Contributors, *Polylogarithms and their Arithmetic Bridges*, canonical manuscript README, repository `VladimirReshetnikov/ProveIt`, path `Analysis/Polylogarithms/docs/manuscript/README.md`. Inspected blob `dc0fa26d0e5c2ede56df37c5640c7319681dfd25`.

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[9] ProveIt Contributors, *Further research questions and concrete next targets*, file `sections/further_research.tex` in the same continuation as [8]. Inspected blob `1051d5c4348e9b697712468e9c6b0c0f4d6373db`.

[10] ProveIt Contributors, *Gamma, polygamma, and Stieltjes identities*, file `sections/03-jets.tex` under `stieltjes-correlation-calculus/continuations/mellin-lerch-dilation`, retrieved at repository commit `64c0a386f5eb0cfc7e1ebb751b46eaed20f3f84a`. Full paths and the scope of the comparison are recorded in `integration/SOURCE_AUDIT.md`.